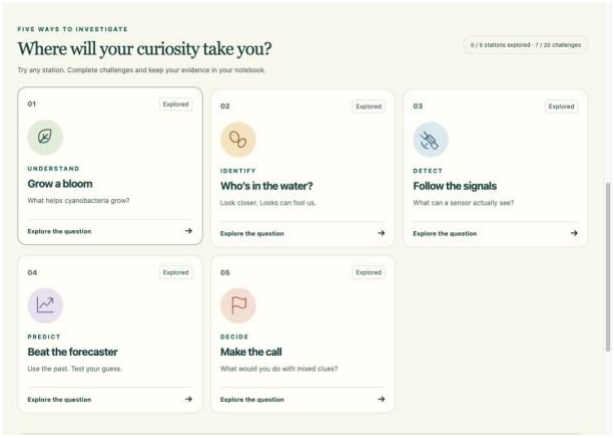
Lake Lab Student Guide

A step by step guide for middle school lake detectives

Use Lake Lab to explore Lady Bird Lake in Austin, Texas. You can change a model, compare pictures, read sensor clues, train a small computer model and make decisions. Then save what you learned for your own project.

You do not need to finish the whole guide at once. Start with pages 2 and 3. After that, choose the station that matches your question.



The five station cards open five different ways to investigate the lake.

Find your next activity

Pages	What you can do
2–3	Get started and learn what counts as evidence
4	Understand growth conditions
5	Identify organisms and teach a classifier
6–8	Detect signals with sensors, pixels and light
9–10	Predict outcomes and make decisions
11–12	Save your work and plan your own investigation
13–14	Find word meanings, help and image credits

The app is a learning tool. It does not tell you whether the lake is safe. Use it on a screen; these activities do not ask you to collect algae or enter the water.

Start your first visit

Open the link your teacher or coach gives you. No account or name is needed. If a link starts with 127.0.0.1, it works only on a computer running the app locally. Ask your teacher for the correct link on your device.

1. Select Explore, then Choose a station. Select a station card.

2. Select Try it to reach the controls. On a phone, scroll down to see the full activity.

3. Change one setting or make one choice. Notice what changes.

4. Open How this works. It explains the model and what it leaves out.

5. Select Take a challenge. Read its goal. Use a setup button if one is shown.

6. After the activity, select Check my evidence on that challenge. A check mark records progress.

7. Select For your project. Write a finding or question, then select Save finding + settings to notebook.

Three different ways to save

Control	What it does
Record this comparison or Record this spectrum	Keeps a comparison inside that station. It does not write your explanation.
Check my evidence	Checks an activity goal and saves its completion. A check mark is not proof that a scientific claim is correct.
Save finding + settings to notebook	Saves your words and the station settings together.

Move around without losing your place

Use All stations or Explore to return home. Use Notebook to see your notes. Use Glossary to find a word. Use Data & sources to find where the evidence came from. Some underlined words also open a short definition.

Keyboard tip: press Tab to move between controls. Use arrow keys on a slider or menu. Press Enter or Space to use a button. A clear outline shows the control you are using.

Know what your evidence means

The local lake is different

Cyanobacteria are tiny living things that use sunlight. Some can make toxins. They are often called blue-green algae, but they are bacteria. A bloom is a large amount of growth.

At Lady Bird Lake, an important problem is mats that grow on the bottom. A benthic mat is a layer of organisms on the lake bed. Pieces can break free and float up. Clear-looking water can still hide a mat.

The lake is a narrow part of the Colorado River held between dams. A sensor in open water may miss the bottom or shore. A large satellite pixel may mix water with land.

Read the badge before making a claim

Badge or word	Plain meaning	A useful sentence
Measured or Real image	An observation or picture from a real place and time. It is not live.	This record shows what was observed then.
Derived	Real records were selected, grouped or calculated.	I need to check how the summary was made.
Simulated or Illustrative	People made a teaching example with chosen rules.	In this model, changing this setting did this.
Provisional	The record still has quality questions.	I would check the flags before trusting this result.

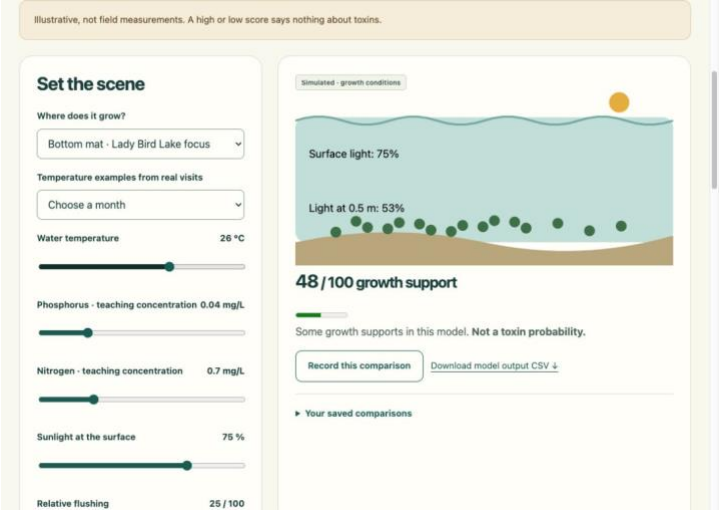
Do not turn a clue into a safety claim

Green colour is not a toxin test. A pigment signal is not a toxin test. A low growth score is not a safety test. A missing result means we do not have that result. It does not mean zero.

Before using a graph, ask: Where was this measured? When? What was measured? What are the units? What is missing?

Understand growth conditions

Open Understand — Grow a bloom. A model is a simplified set of rules. Here, the drawing and score change when you move the controls.



Growth controls are on the left. The lake drawing and growth-support score are on the right.

1. Select Reset scene. Keep Bottom mat selected.

2. Set Water temperature to 15°C. Select Record this comparison.

3. Set only Water temperature to 28°C. Record again. Compare the scores.

4. Switch Where does it grow to Floating bloom. Compare it with the mat at the same settings.

5. Open Take a challenge. Try Follow the warmth. Save one finding under For your project.

What the controls mean

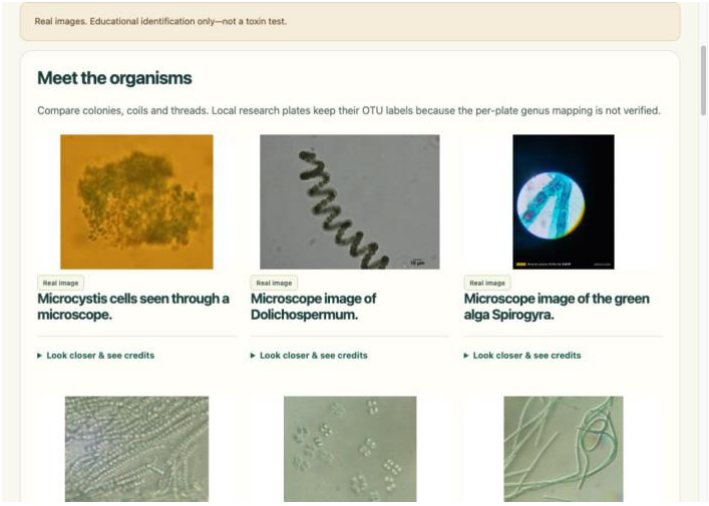
Temperature tells how warm the water is. Nitrogen and phosphorus are nutrients organisms use to grow. Their unit, mg/L, means milligrams per litre. Sunlight supplies energy. Relative flushing describes how strongly water is being replaced in this example. It is not a measured river-flow value.

The score is growth support out of 100. It is not the amount of algae, a toxin level or a percent chance of danger. Very hot water does not keep raising the score: this model has a preferred temperature.

Try next: Load October and change other controls. The temperature example comes from real visits, but the other preset settings are invented. Ask why a month alone cannot tell you what will grow.

Identify what is in the water

Open Identify — Who's in the water? Compare shapes first. Microcystis can form groups of cells. Dolichospermum forms chains. Spirogyra is a green alga. Duckweed is a small plant, not a cyanobacterium.



Open Look closer & see credits to read the source and limits of an image.

1. Open the details on at least three organism cards. Look for groups, threads or coils.

2. Scroll to Teach the machine. Choose labels for at least three training images.

3. Select Train nearest-neighbour model. Read both test predictions and their source labels.

4. Change one training label on purpose. Train again. Did either prediction change?

5. Scroll to Check a field photo. Answer each question and use Next photo question.

What the machine is doing

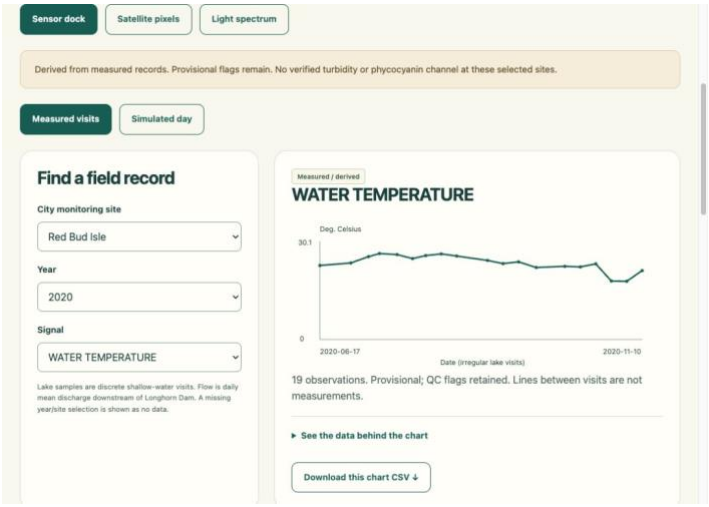
A classifier sorts examples into groups. This one compares colour and edge contrast. It copies the label of the closest training image. It does not understand cells or toxins.

All four training images in this small collection are cyanobacteria examples. The two test images include one cyanobacterium and one green alga. A score on only two pictures is weak evidence. One answer changes it by 50 percentage points!

OTU is a research group label on the local image plates. The app keeps those labels because the exact genus names for each plate still need review. A wrong label is still wrong even if the tiny test score improves.

Detect changes with sensor clues

Open Detect — Follow the signals, then Sensor dock. A sensor measures one kind of signal at one place. Its location matters.



Measured visits show dated field observations. Open the data table to inspect the values and flags.

1. Select Measured visits. Try Red Bud Isle, year 2020 and WATER TEMPERATURE.

2. Read the dates along the bottom and units at the side. Open See the data behind the chart.

3. Change Signal to PH or DISSOLVED OXYGEN. Keep the site and year the same.

4. Select Simulated day. Compare pH, dissolved oxygen and Phycocyanin signal.

5. Choose which pair rises before the pigment pulse. Select Check this comparison.

Read each signal carefully

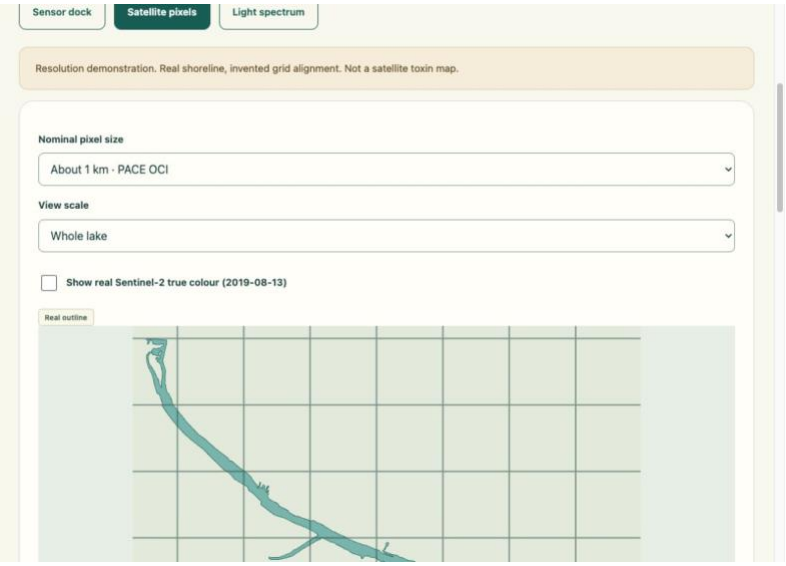
pH describes how acidic or basic water is. Dissolved oxygen is oxygen in the water. Turbidity describes cloudiness. A phycocyanin probe looks for a signal from a blue pigment. None of these readings directly measures toxins.

The real lake records are separate visits, not a continuous recording through a day. Lines between visits only connect the points. Downstream river flow comes from below Longhorn Dam, not from the lake itself.

The simulated day is invented. Its order of changes is not a rule for every lake. RFU means relative fluorescence units: a signal tied to an instrument. It is not automatically a pigment concentration.

Detect what satellite pixels miss

In Detect, select Satellite pixels. A pixel is one small area represented in an image. A mixed pixel includes more than one surface, such as land and water.



The same real lake outline is covered by a demonstration grid. This screen shows cells about 1 km wide.

1. Under Nominal pixel size, select 10 m · Sentinel-2 visible bands. Try Zoom in 4× if the lines are hard to see.

2. Select 300 m · Sentinel-3 OLCI, then About 1 km · PACE OCI. Keep the same view scale while comparing.

3. Find cells that contain both land and lake. Does a bigger cell make the lake wider?

4. Turn on Show real Sentinel-2 true colour to compare the outline with an image from August 13, 2019.

5. Return to the challenges and compare 10 m with 1,000 m pixels.

What you can conclude

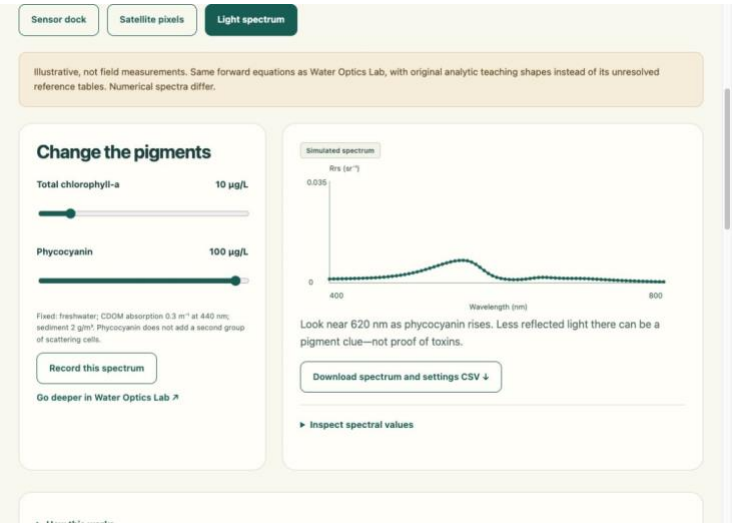
Bigger cells can mix more land into a narrow lake. The lake has not changed; the view has changed. A 1 km cell is about 1,000 m by 1,000 m.

This grid is a size demonstration, not the exact position of real satellite pixels. The colour image is a real old picture, not a current toxin map. A bottom mat may be hidden from a satellite.

For your project: what size of feature would your idea need to notice? Where would the sensor need to look?

Detect a pigment clue in light

In Detect, select Light spectrum. A spectrum separates light by wavelength. Wavelength is a way to name a colour of light; nm means nanometres.



The horizontal axis shows wavelength. Rrs on the vertical axis describes reflected light relative to incoming light.

1. Keep Total chlorophyll-a at 10 µg/L. Chlorophyll-a is a pigment used in photosynthesis.

2. Set Phycocyanin to 0 µg/L. Select Record this spectrum.

3. Set Phycocyanin to 30 µg/L or more. Keep chlorophyll unchanged. Record again.

4. Look near 620 nm. Open Inspect spectral values to compare numbers.

5. Select Download spectrum and settings CSV if you want a table for your project notes.

Why the curve changes

Phycocyanin, or PC, is a blue pigment found in many cyanobacteria. In this model it absorbs light near 620 nm. More absorption can mean less light comes back from the water at that wavelength.

µg/L means micrograms per litre, a unit for small concentrations. You do not need to calculate Rrs to use this activity. Compare the shape and the values at the same wavelength.

The spectrum is simulated. It uses the Water Optics Lab equations with simpler teaching inputs, so its numbers differ from that app. It does not include bottom-mat reflections or toxins. Sentinel-2 does not have a narrow 620 nm band: seeing a model curve is not the same as measuring it with that satellite.

Predict and test a forecast

Open Predict — Beat the forecaster. Start in Simulated training game. Every visit and event in this game is invented.

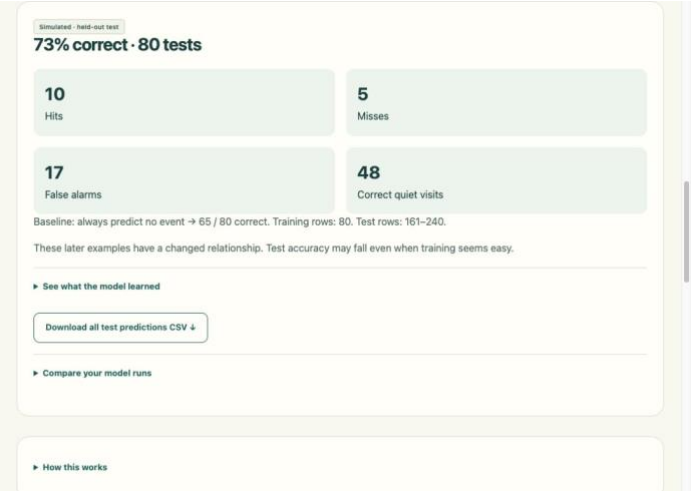
1. Read the four clues and select I predict an event or I predict no event. Read the reveal.

2. Under What can your model use, select at least one input. An input is a clue given to the model.

3. Set Early training examples to 80. Select Train and test my model.

4. Record the hits, misses, false alarms and correct quiet visits. Compare with the baseline.

5. Change one input choice or the number of training examples. Train again. Open Compare your model runs.



A test result for one chosen model. The baseline can score better by always predicting the common outcome.

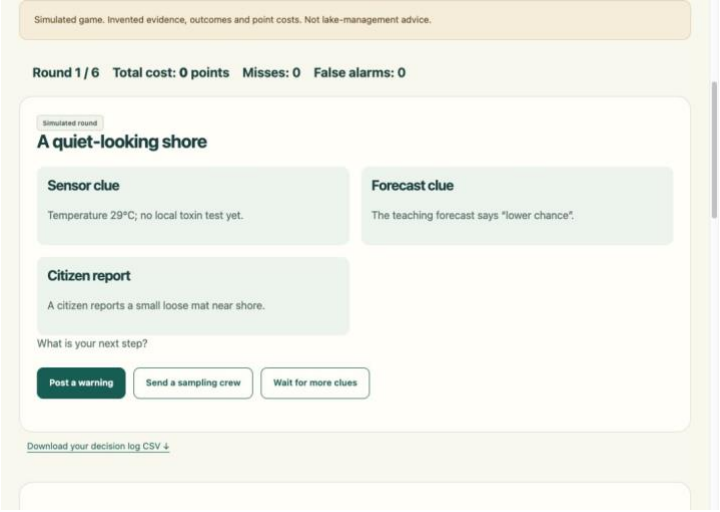
Training examples are practice data. Held-out examples are kept out of that practice. The last 80 visits are the test set. A baseline is a simple comparison rule, such as always guessing no event.

A hit finds an event. A miss overlooks one. A false alarm predicts an event that does not happen. A correct quiet visit predicts no event and gets it right. A high percent correct can hide many misses.

Try Real evidence explorer next. Select Reveal next sample records. These are real historical groups of samples, not safe/unsafe answers. Mat and water samples use different units. A below-limit result is not proof of safety.

Decide what to do with mixed clues

Open Decide — Make the call. You are choosing actions in a six-round game. Read the sensor clue, forecast clue and citizen report before each choice.



Choose an action before the hidden result is revealed. Every round and its point costs are invented.

1. Choose Post a warning, Send a sampling crew or Wait for more clues. Explain your reason first.

2. If you send a crew, read the invented test result. Then choose warning or waiting.

3. Read the reveal and cost. Select Next round. Finish all six rounds.

4. Download your decision log or save a notebook finding before selecting Replay. Replay starts a new log.

5. Try a different strategy. Compare misses and false alarms, not just the total cost.

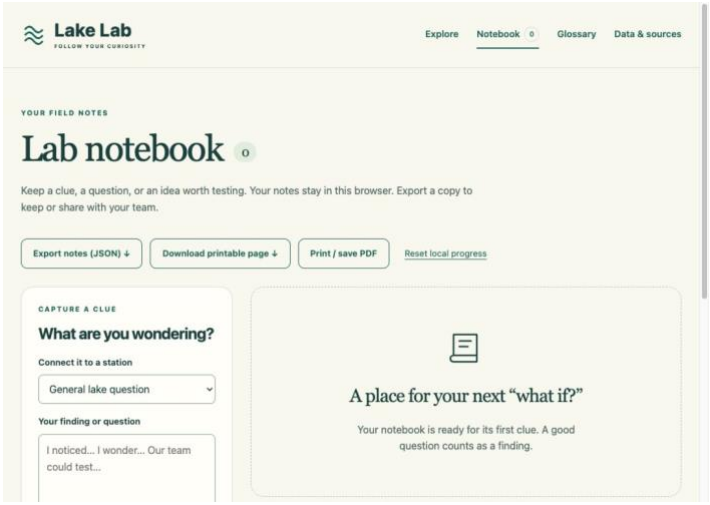
Choice	Event happens	No event happens
Warn	2 points	5 points
Wait	12 points	0 points
Sample first	Add 3 points, then choose	Add 3 points, then choose

Lower cost is better under these game rules. The points are not money or an official policy. Sampling is instant and perfect only in the game. Real tests take time and can miss things.

For your project: whose needs matter when evidence is unclear? What new evidence would change your next action?

Save a notebook you can explain

Save your thinking, not just a score. At the bottom of a station, use For your project. This button saves the station settings with your note. A general note written directly in Notebook does not capture a station experiment.



Notebook lets you write, edit and export notes. Reset local progress is a separate action that clears local work.

Try this four sentence note

I changed __ while keeping __ the same. I noticed __. This evidence is real / derived / simulated because __. It cannot tell me __, so my next question is __.

Keep a copy

1. Open Notebook. Read a saved note and its Saved experiment settings. Use Edit note if your explanation needs work.

2. Select Download printable page. Open the downloaded HTML file in a browser. Print it or save it as PDF. You can also try Print / save PDF directly in the app.

3. Use Export notes (JSON) for a structured copy of the saved work. JSON is a computer-readable text file, not a normal report.

4. Download chart CSV files separately if you need the plotted numbers. CSV files open as tables in spreadsheet software.

There is no account, automatic sharing or device sync. The current app has no button to import a JSON backup. Keep your printable copy and data files. Do not put names, addresses or other personal details into notes.

Export before clearing browser data, changing devices or using Reset local progress. If storage is blocked, notes may last only while the page stays open.

Plan your own lake investigation

Choose a question your team cares about. The app gives you ways to think and test. Your team decides what its own solution should be.

Choose one route

Your question	A useful route
What might support growth?	Understand → Detect sensor dock
Could a camera be fooled?	Identify → Detect satellite pixels
Would a forecast help?	Predict → Decide
What evidence should our idea collect?	Detect → Notebook → Data & sources

Write a plan before your next visit

Our question is _____

One setting or choice we will change is _____

Things we will keep the same are _____

Evidence we will save is _____

A result that would change our minds is _____

A limit of this test is _____

Bring something useful to your team

- A notebook finding that says whether the evidence is simulated or real.
- A table or image with its date, source and units when available.
- One possible explanation and one different explanation.
- One question you still cannot answer.

You can revisit completed challenges. A saved check mark shows that you met an activity goal. It does not mean you have finished investigating the lake.

Words and help

Word	Meaning
Benthic	On or near the bottom of a body of water.
Bloom	A large amount of algae or cyanobacteria growth.
Toxin	A harmful substance made by a living thing.
Pigment	A substance that absorbs some colours of light.
Nutrient	A material organisms need to grow, such as nitrogen or phosphorus.
Model	A simplified set of rules used to explore or predict something.
Classifier	A model that sorts examples into groups.
Held out	Kept separate from the examples used to train a model.
Baseline	A simple rule used for comparison.
Pixel	One cell of an image; it can include both land and water.
Uncertainty	What we do not know, or how unsure a result is.
Below limit	The reported result is below a test threshold; not proof of zero.

If something does not work

- No graph values: check the site, year and signal. Some combinations have no observations. Try Red Bud Isle, 2020, WATER TEMPERATURE.
- Challenge will not pass: read the exact goal, use its setup button and record the requested comparison. Then select Check my evidence.
- No model result: choose labels in Identify or at least one input in Predict, then train.
- A note disappeared: check the device and browser. Ask your teacher about storage limits. Use exports next time.
- Small screen: scroll down or use browser zoom. The graph also has a data table.

Where the evidence comes from

Open Data & sources in the app. Each collection has methods, limits and downloads. The source records are frozen copies. They do not update while you use Lake Lab.

Source	What it contributes
City of Austin	Historical water-quality visits and available toxin records.
USGS	River flow below Longhorn Dam and a lake outline.
NOAA	Austin Camp Mabry air temperature and rainfall.
Perri and colleagues	Images and research from the local 2019 event.
Copernicus Sentinel-2	An old true-colour satellite image.
Lake Lab teaching models	Invented growth scores, daily sensor patterns, training events and manager rounds.

Give credit when you share

Say “Lake Lab teaching simulation” for a model result. Say which source, date and place a real observation comes from. Keep the source and licence with an image. Do not describe a screenshot as your own field measurement.

Lake Lab was created by Chintan B. Maniyar. Guide text and original app illustrations: CC BY 4.0. Screenshots show the local app on October 10, 2026. Some counters and settings reflect demonstration use. Your screen may differ. Third-party images keep their own licences.

Microscope images visible in the Identify screenshot: Microcystis by Panek, CC BY 2.5; Dolichospermum by Masa Zupancic, CC BY-SA 4.0; Spirogyra by SPJ777, CC BY 4.0. Local OTU plates: Katherine Perri, Schonna Manning, Matt Ashworth and Brent Bellinger, CC BY 4.0. Images are shown within a screenshot and resized for the page; labels and colours were not retouched.

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More to explore: Water Optics Lab at chintan2108.github.io/Water_Optics_Rrs_Simulator/ . Ask your teacher before following outside links. Lake Lab creator contact: chintanmaniyar@gmail.com.